

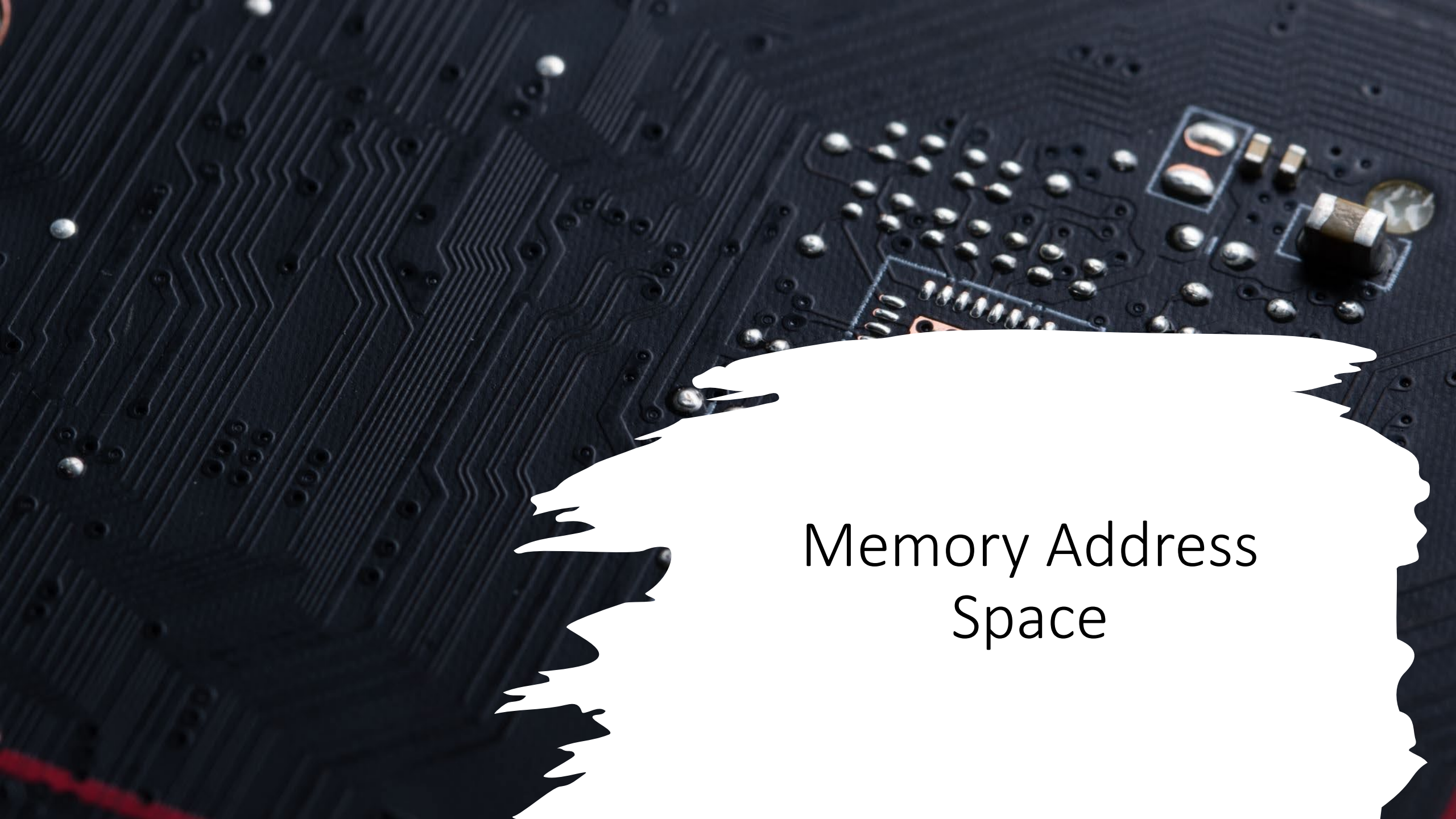


SC1006

Computer Organisation and Architecture

Virtual Memory

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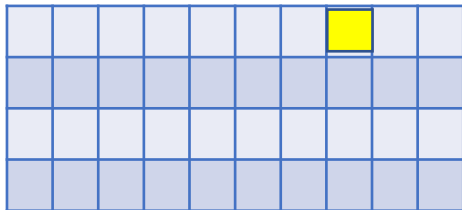
Memory Address
Space

Memory Address Space (Analogy)

Address Space 3
(Employee)



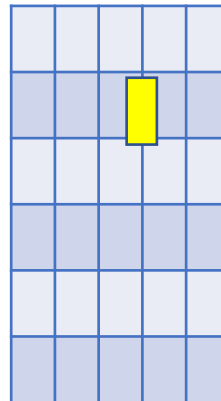
Cabinet (Company Name)



Address Space 2
(Manager)



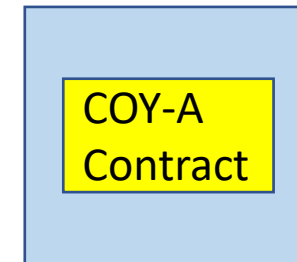
Cabinet (Market Segment)



Address Space 1
(Boss)



Drawer (all-in-one)

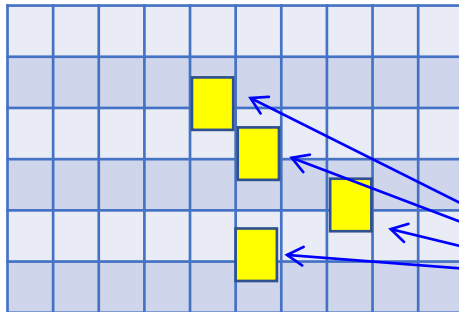


Memory Address Space (Computing)

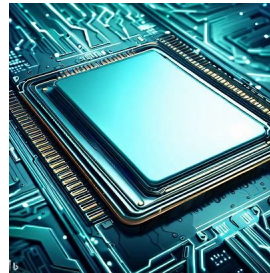
Address Space 3
(Storage Memory
Controller)



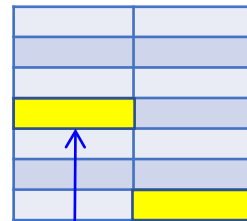
Storage Memory
Addressing (LBA-
Logical block addressing)



Address Space 2
(System Memory
Controller)



System Memory
Addressing



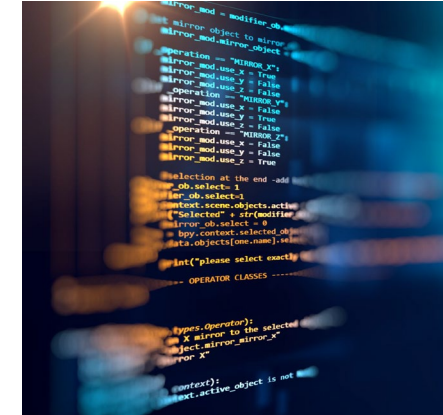
Test Video

OS



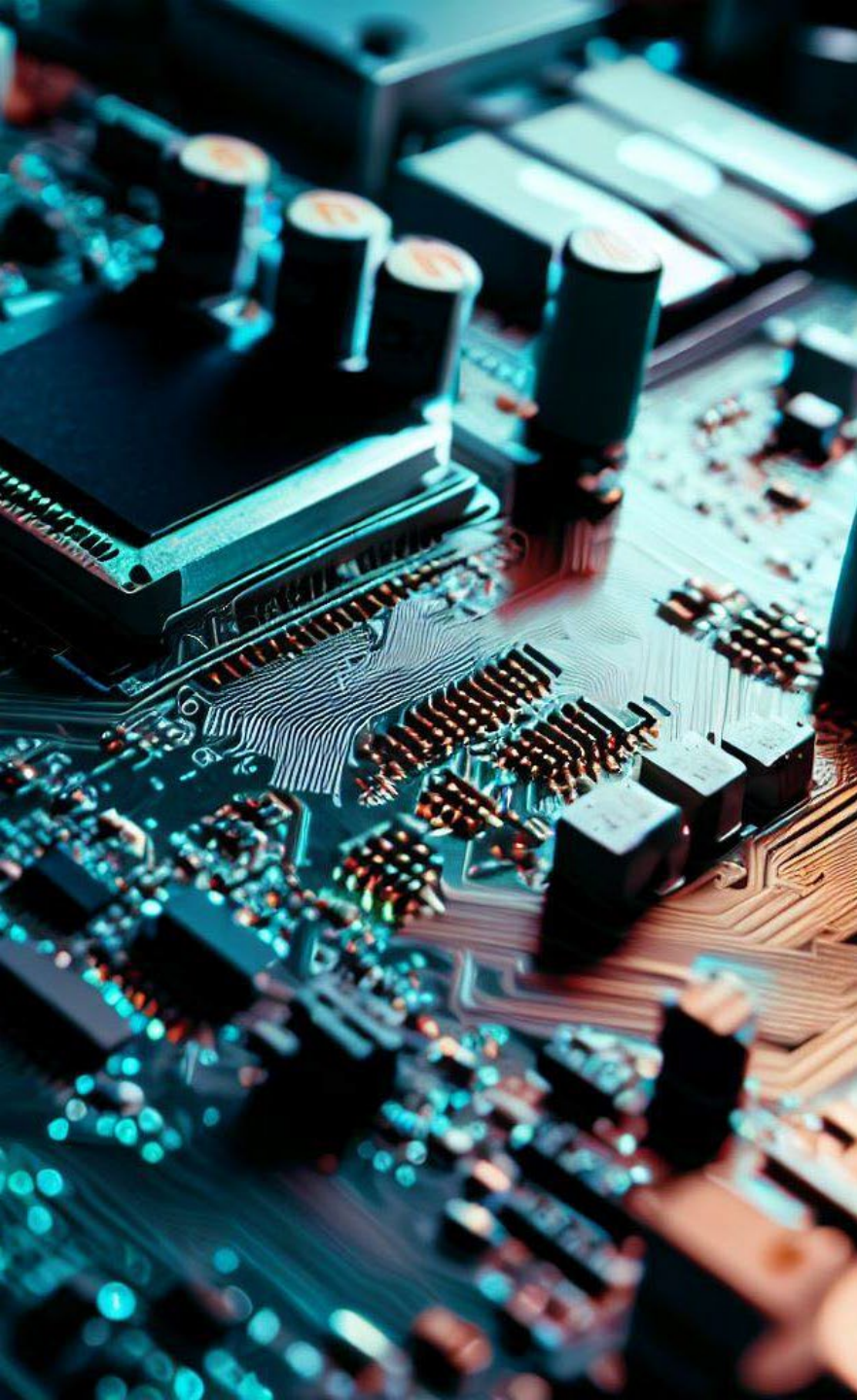
OS keeps track of
the location of
Test Video in the
System and
Storage memory

Address Space 1
(Program)



Virtual Memory
Addressing





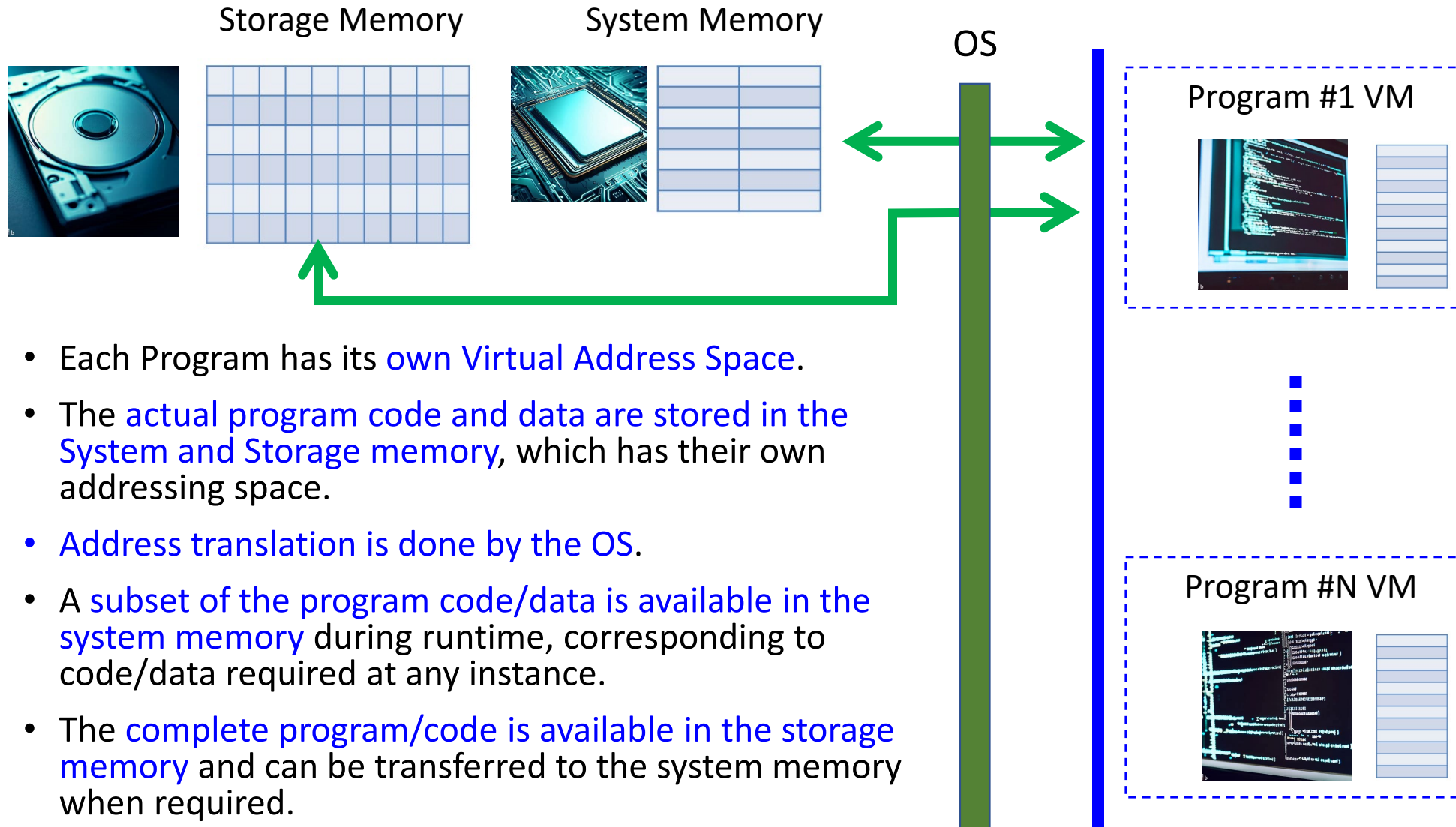
Basic Concepts

- In computing, address space is a range of discrete addresses corresponding to some physical or logical entity, e.g. program virtual memory, physical system memory, logical layout of a HDD etc.
- Every piece of data/code in a program is assigned an address by the compiler during the program compilation.
- When executing a program, the information it requires is obtained by issuing a request to the operating system (OS) using the addresses assigned by the compiler.
- The program doesn't need to know how the required information is organised in the system memory and rely on a middle man to do the corresponding address translation in order to fetch the correct information.
- In this case, the operating system is the middle man, translating the compiler generated address to the system memory address where the required information is stored.
- Program will still work as long as it gets the code/data it requested.



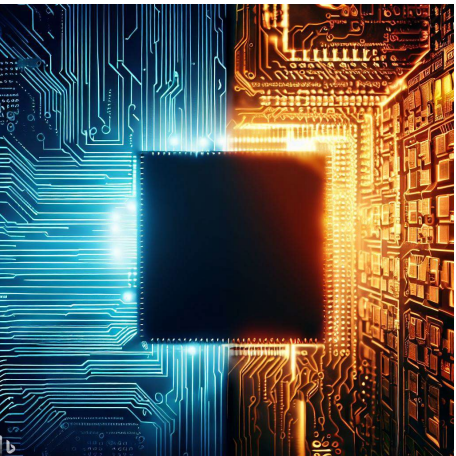
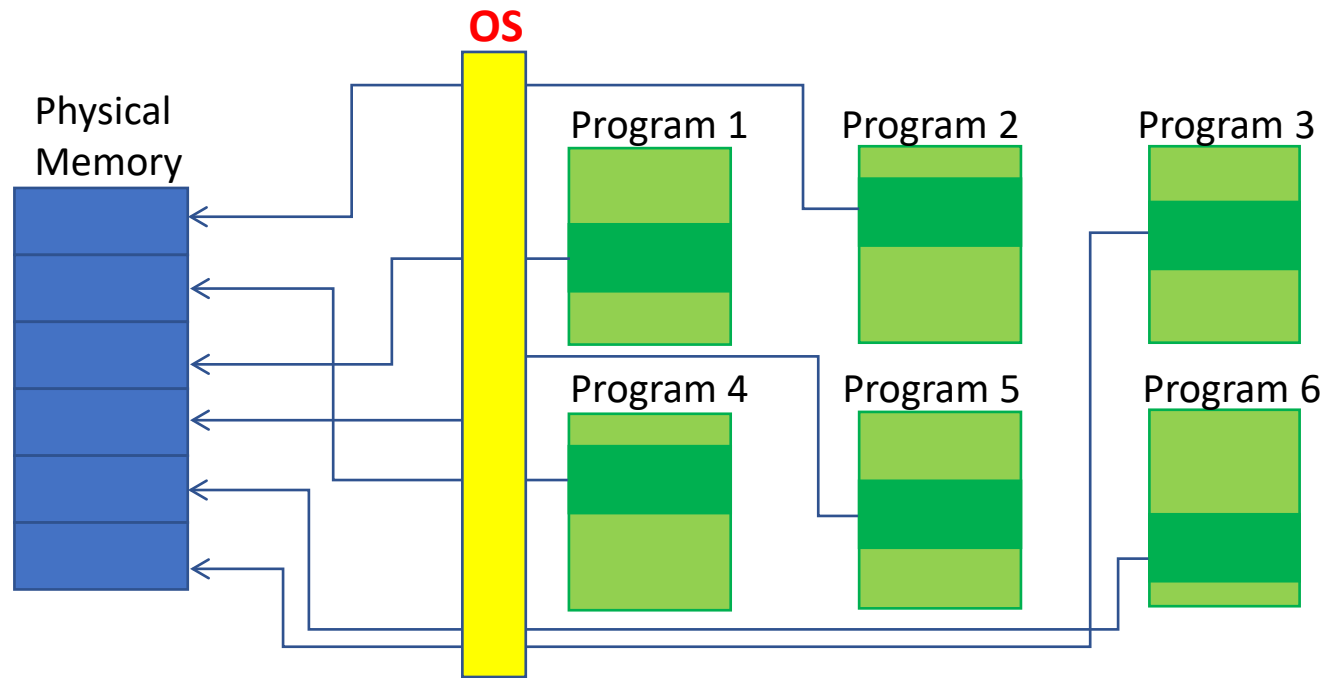
Virtual Memory Management

Virtual Memory (VM) Management



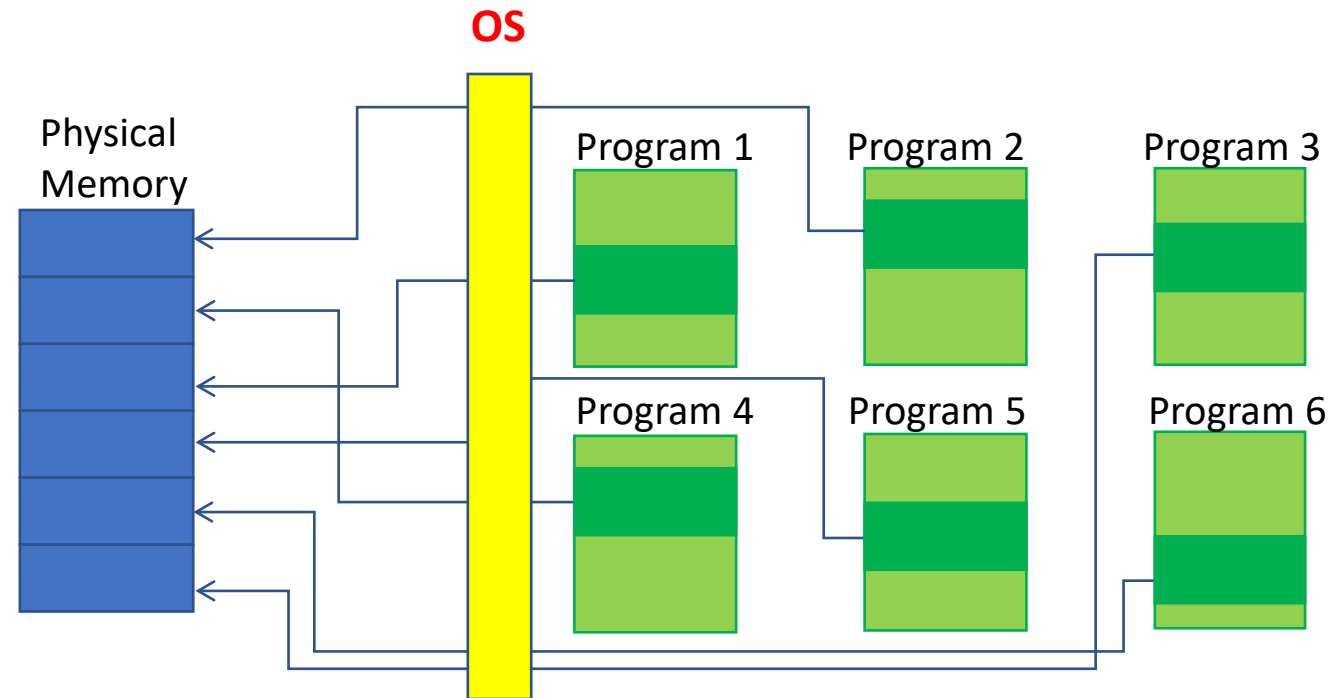
Virtual vs Physical Memory Address Space

- The addresses used by the program is generated by the compiler and is known as the **virtual address**.
- The virtual address of the code/data is typically **different from the Physical Memory Address** that they will be resided.
- **Address Translation** is thus required and is done in hardware and/or software, managed by the **Operating System (OS)**.
- Note that Physical, System and Main memory refers to the same memory for this course.



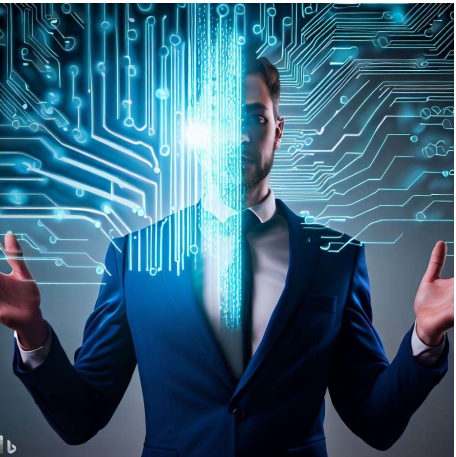
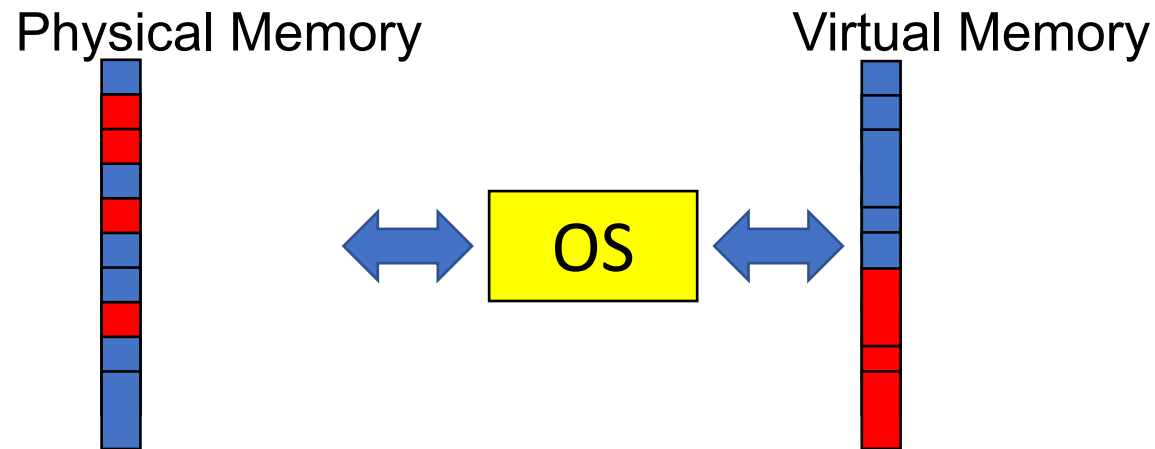
Advantages of Virtual Memory

- OS is able to isolate each virtual memory space and prevent corruption between these spaces.
- Allow efficient and safe sharing among different programs within the shared physical memory.
- Allow one or more programs to run in the physical memory **simultaneously** even if the **total size** of all the programs is **larger** than the **actual physical memory** size.



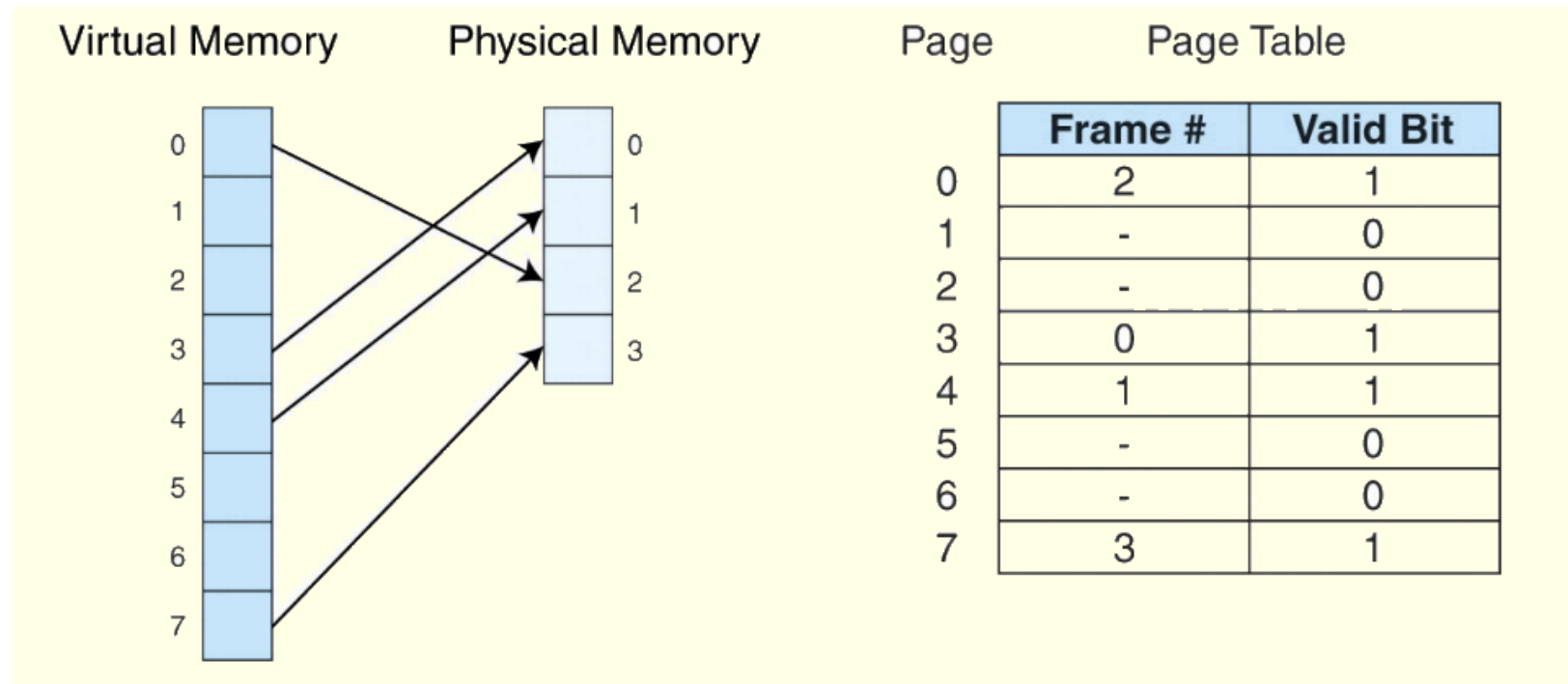
Advantages of Virtual Memory

- OS is able to relocate virtual memory blocks to any physical memory blocks. This allows the virtual address space seen by the application to appear to be contiguous when it is actually spread across fragmented blocks in the physical memory.



Address Mapping

- There are two common schemes used in the industry for address mapping
 - **Paging**. Memory space partitioned into Fixed sized blocks
 - **Segmentation**. Memory space partitioned into variable sized segments.
- For our course, we will only deal with **paging with single level page table**.

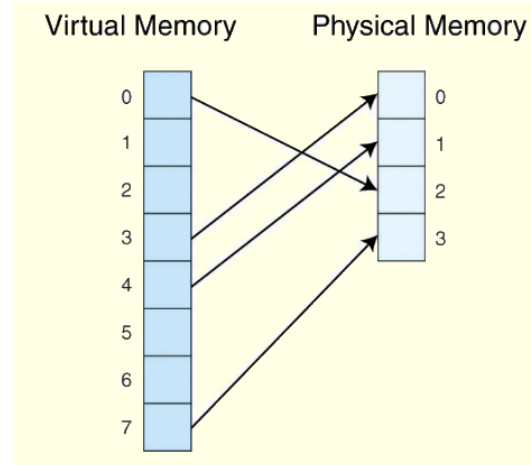
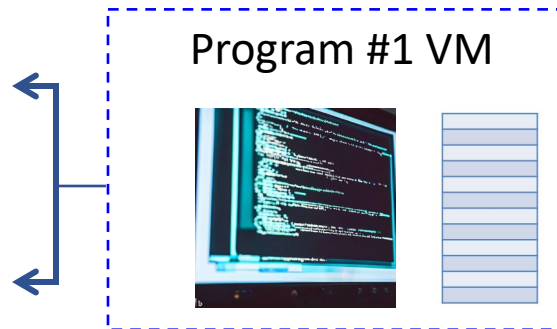
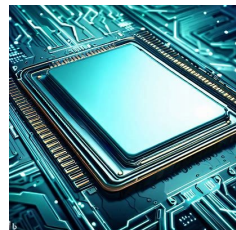


Source: Linda Null



Paging Method

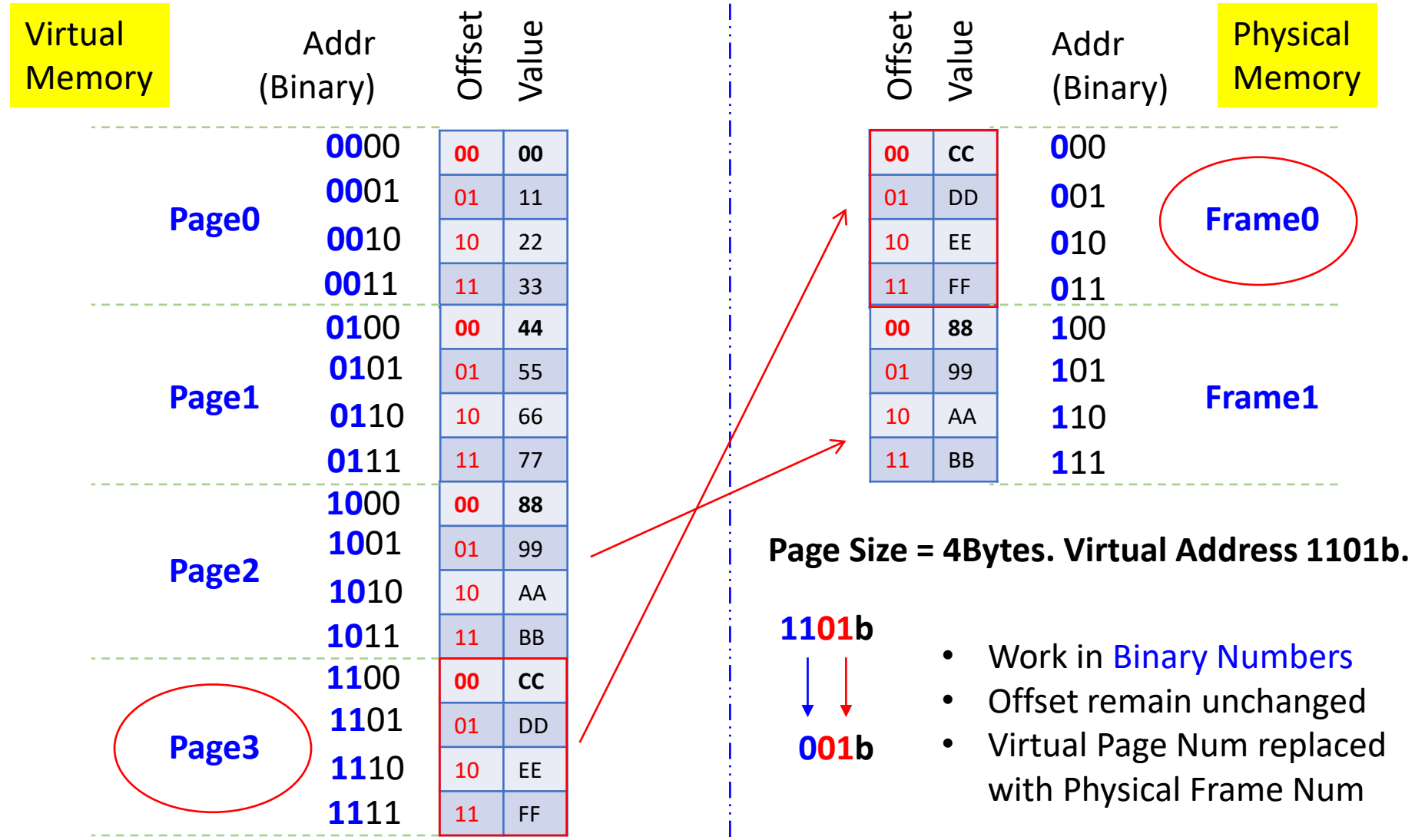
- In a system that uses paging, the memory space is partitioned into **fixed size blocks** known as a **Page/Frame**.
- **Information concerning the location of each page**, whether on disk or in memory, is maintained in a data structure called a **page table** (shown below).
- In the Page Table below
 - **Frame** refers to the **physical frame** number in the main memory.
 - **Page** refers to the **virtual page** number used by program code.
 - **Valid Bit (VB)** indicates whether the Virtual Page is in the main memory (VB=1) or not (VB=0).



Page	Page Table	
	Frame #	Valid Bit
0	2	1
1	-	0
2	-	0
3	0	1
4	1	1
5	-	0
6	-	0
7	3	1

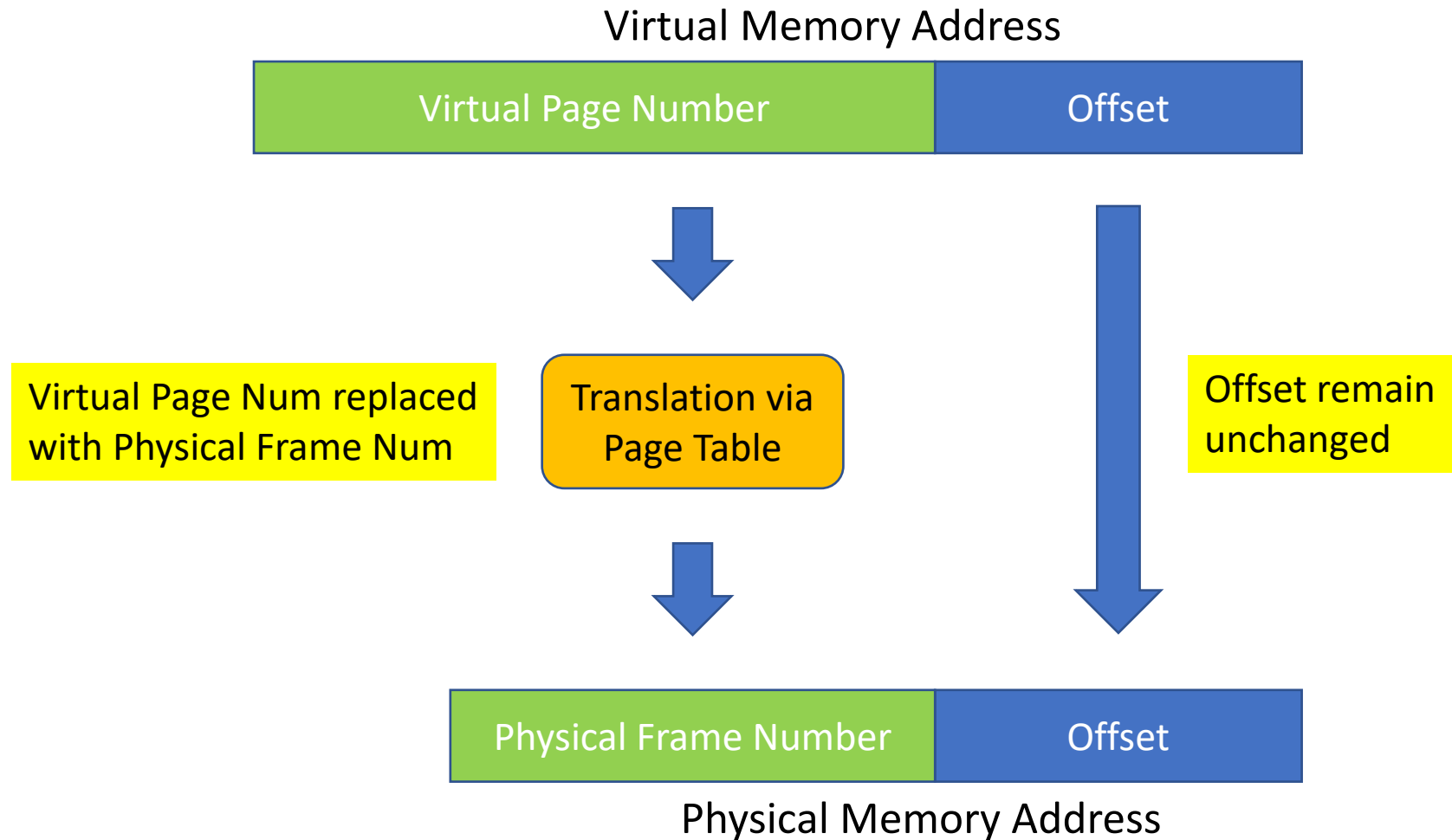
Source: Linda Null

Address Translation



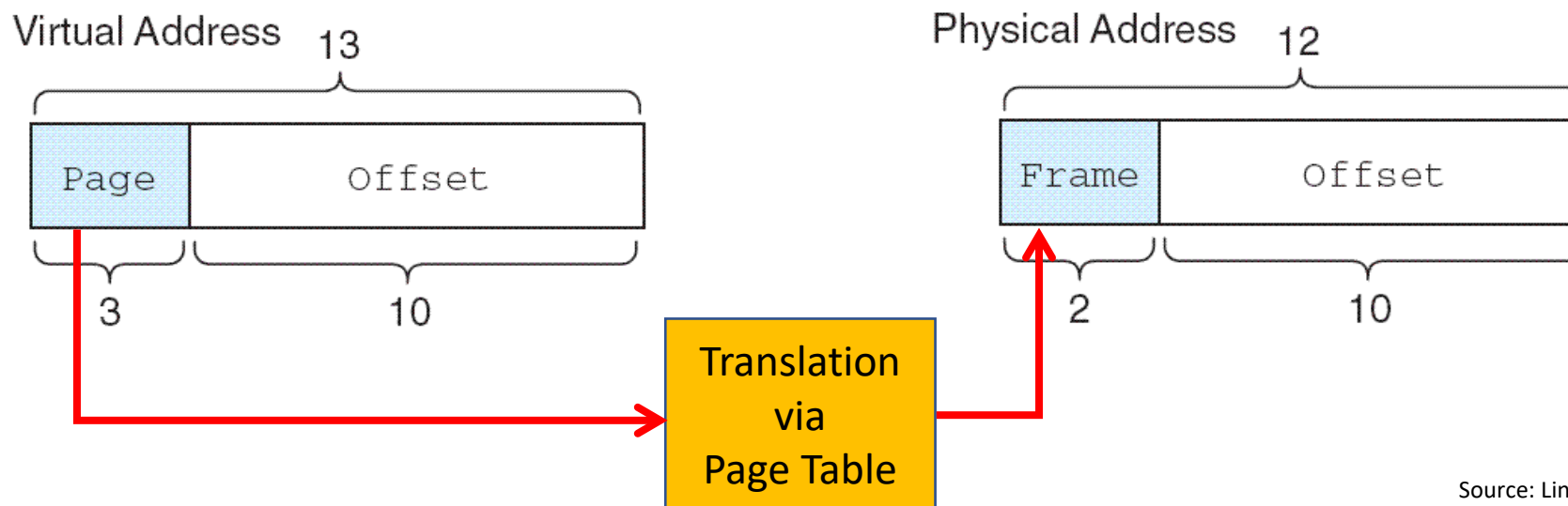
Address Translation

Work in Binary Numbers format



Paging Example

- Consider a system with a **virtual address space of 8K** and a **physical address space of 4K**, and the system uses byte addressing.
- If **page size is 1KByte**, we have 8 virtual pages mapping to 4 physical frames.
- Note that **Virtual Page Size is always equal to Physical Frame Size**.
- A virtual address has 13 bits ($8K = 2^{13}$) with 10 for the offset, because the page size is 1024, and 3 bits for the page field.
- A physical memory address requires 12 bits, the first two bits for the page frame and the trailing 10 bits the offset.



Paging Example

- Suppose we have the page table shown below.
- What happened when CPU access virtual address location
 - 0x1553
 - 0x0FA0
- 0x1553 = 1010101010011b
 - Data resides in Virtual Page 5
 - From Page Table, Virtual Page 5 is mapped to Physical Frame 1 and Valid bit is 1.
 - Physical Address = 010101010011b = 0x0553
- 0x0FA0 = 0111110100000b
 - Data resides in Virtual Page 3
 - From Page Table, Valid bit is 0
 - Data not in Physical Memory
 - **Page Fault** (triggers the OS to load the required page from storage memory to the Physical Memory)

Page Table		
Page	Frame	Valid Bit
0	-	0
1	3	1
2	0	1
3	-	0
4	-	0
5	1	1
6	2	1
7	-	0

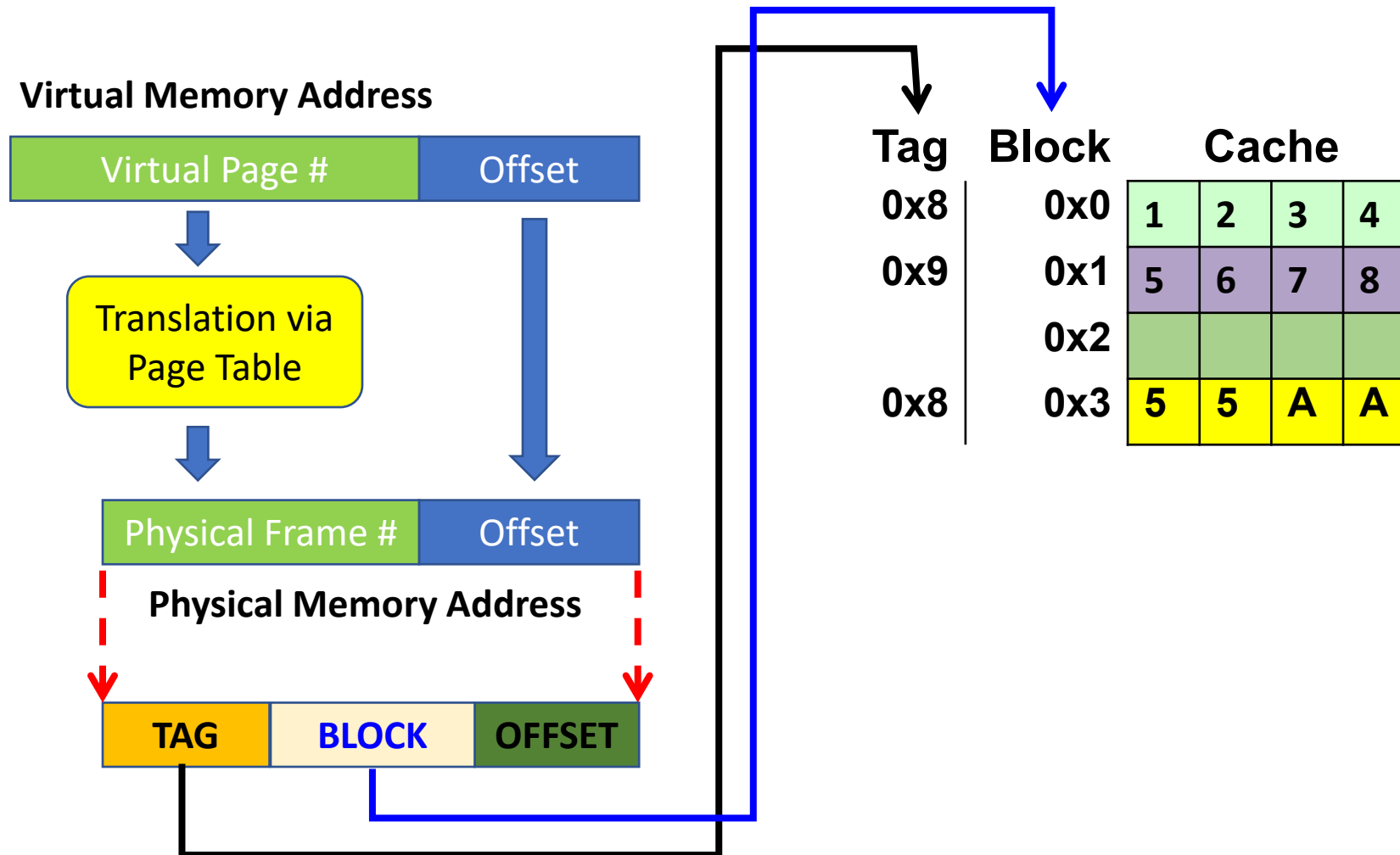
Paging Example



- In the previous slide, when accessing virtual address 0x1553 (which is in virtual page 5), the page table shows the valid bit is 1. From the page table, physical frame 1 will be used for subsequent address translation needed to access the actual data.
- When accessing virtual address 0xFA0 (virtual page 3), the page table shows that the valid bit is zero. If the valid bit is zero in the page table entry for the virtual address, this means that the page is not in main memory and must be fetched from Storage Memory.

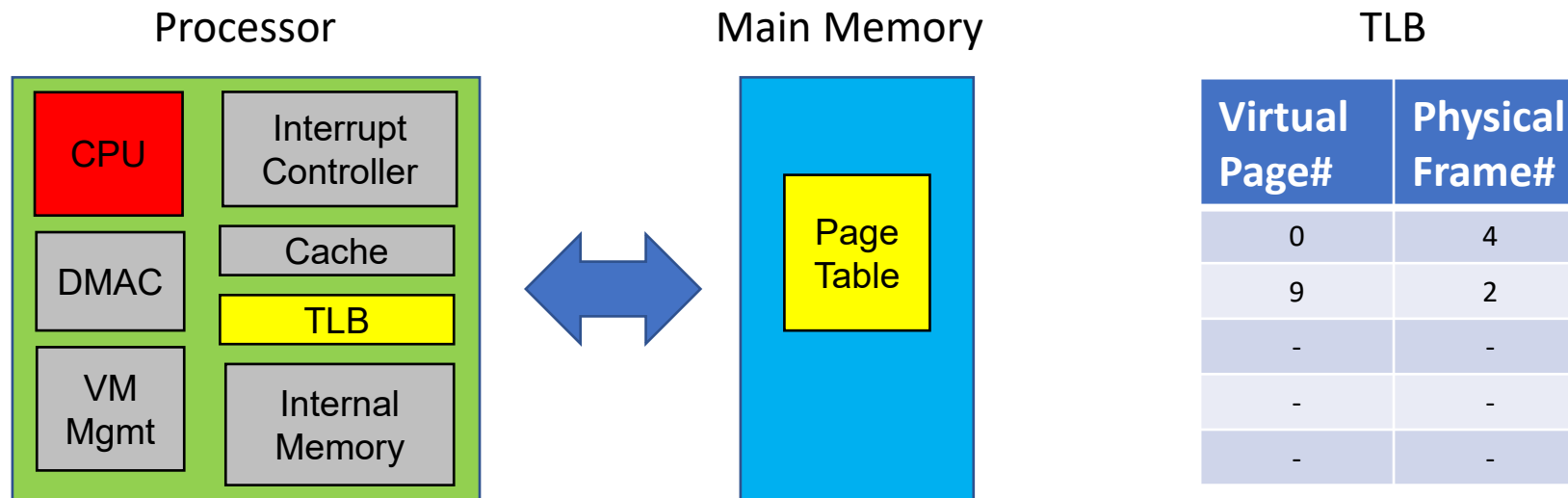
Integrating Cache and Virtual Memory

- Assumption: Cache uses Physical Memory Address to perform Mapping.

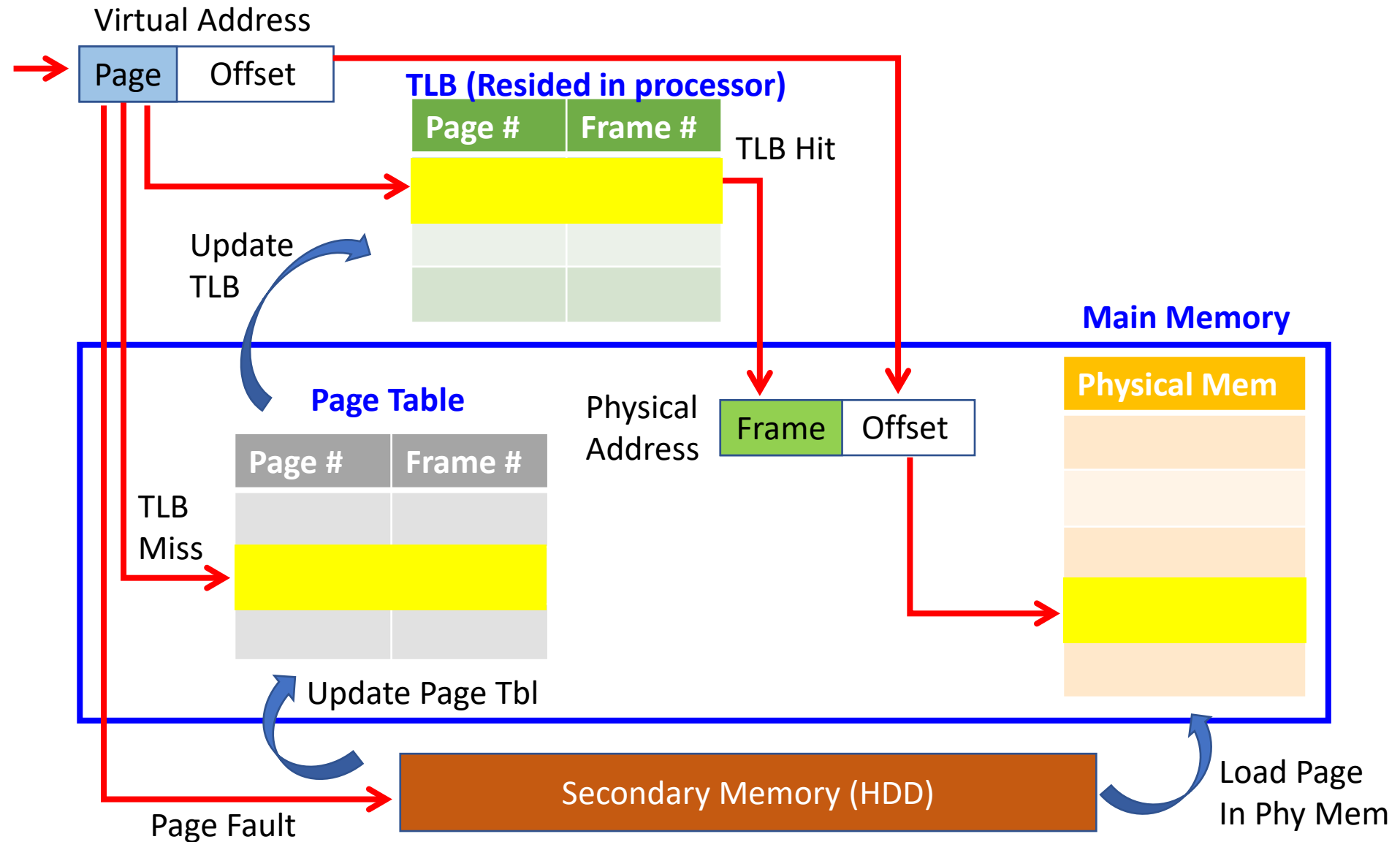


Translation Lookaside Buffer (TLB)

- **Page Table** is located in **Main Memory** so access is **slower** than internal memory access.
- To **speed up** the page translation process, a **page table cache (TLB)** is used to store a list of most recently/frequently used address translation entries in the page table, as a **subset** of Page Table.
- TLB is implemented with **fast memory** such as SRAM and reside **within the processor**.
- Each **TLB entry** includes a **Virtual Page number** and its corresponding **Frame Number**.



TLB Lookup Process



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